MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8171

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Tetrode

See figure 1

Mounting position: Vertical, base down or up

Weight: 12.2 pounds nominal

ABSOLUTE RATINGS:

Parameter: Unit:	Ef Vac	Eb kVdc	Ec2 kVdc	Ec1 Vdc	Ib Adc	Pg1 W	Pg2 W	Pp kW	T(anode core & seal) °C	Cooling --- Note 1
Maximum:										
Class C Teleg:										
up to 30 MHz:	7.5±5%	7.5	1.5	-500	3.0	75	250	10.0	250	---
30 to 60 MHz:	7.5±5%	7.0	1.5	-500	2.8	75	250	10.0	250	---
60 to 110 MHz:	7.5±5%	6.5	1.5	-500	2.6	75	250	10.0	250	---
Class C Telep: (Anode mod)	7.5±5%	5.0 Note 2	1.0	-500	2.5	75	250	6.65	250	---
Class AB:	7.5±5%	7.5	1.5	---	4.0	75	250	12.0	250	---
TEST CONDITIONS:	7.5	2.0	0.75	Adj	1.0	---	---	---	---	Note 3

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1231	Peak emission	eb = ec2 = ec1 = 2.5 kv	0.65	II	is	53	---	a
1301	Filament current		0.65	II	If	73.0	78.0	Aac
1261	Electrode voltage (grid)		0.65	II	Ec1	-95.0	-127.0	Vdc
1266	Total grid current		0.65	II	Ic1	---	-20	μAdc
1256	Electrode current (screen)		0.65	II	Ic2	---	+15.0	mAdc
1266	Primary-grid emission (control)	Ic1 = 600 mAdc; t = 15; Ef = 9.0 Vac; anode and screen-grid grounded	0.65	II	Isg1	---	-35	μAdc
1266	Primary-grid emission (screen)	Ec1 = 0; t = 15; Ic2 = 550 mAdc; Ef = 9.0 Vac; anode grounded	0.65	II	Isg2	---	-100	μAdc
	Quality conformance inspection, part 2							
1372	Current division (method B, short pulse)	Eb = Ec2 = 1,500 Vdc; Ec1 = -600 Vdc; egk/Ib = 11a	---	---	egk ic2	--- ---	0 1.25	v a
1331	Direct-interelectrode capacitance	Grounded cathode	---	---	Cin Cout Cgp	108.0 18.0 ---	122.0 23.0 1.0	pF pF pF
1331	Direct-interelectrode capacitance	Grounded grid	---	---	Cin Cout Cpk	48.0 18.0 ---	58.0 23.0 0.16	pF pF pF
	Quality conformance inspection, part 3							
---	Service-life guarantee	See note 4	---	---	---	---	---	---
	Periodic-check tests							
2214	RF useful output power	Class AB amplifier; F = 1 MHz (min); Eb = 7.5 kVdc; Ec2 = 1.5 kVdc; Ec1/Ib = 0.52 Adc; Egi/ib = 3.42 Adc	---	---	Po	14.5	---	kW
1042	Shock, specified pulse	No voltages applied; accel = 15 G peak (min); shock = 11 ms half-sine; total impacts = 6 (3 each X and Z axes)	---	---	---	---	---	---
---	Shock, specified pulse end points:							
1261	Electrode voltage (grid)		---	---	Ec1	-95.0	-127.0	Vdc
1266	Total grid current		---	---	Ic1	---	-25	μAdc
1301	Filament current	See note 2	---	---	ΔIf	---	1.5	Aac

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reprocessed to become a source of raw materials, as distinguished from virgin

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Periodic-check tests -Continued							
1032	Vibration, mechanical	No voltages applied; accel = 2 G peak (min); F = 10 to 50 Hz, ascending only; sweep t = 3 to 8 minutes; 1-sweep each X and Z axes	---	---	---	---	---	---
---	Vibration, mechanical end points:							
1261	Electrode voltage (grid)		---	---	Ec1	-95.0	-127.0	Vdc
1266	Total grid current		---	---	Ic1	---	-25	μAdc
1301	Filament current	See note 2	---	---	ΔIf	---	1.5	Aac

NOTES:

1. Minimum airflow requirements for incoming air at 50°C maximum at sea level are shown in table I. In all cases of operation a socket which provides for forced-air cooling of the base shall be used and maximum temperature ratings shall not be exceeded. Ratings apply for bias voltage of less than -500 volts and frequencies less than 110 MHz. Air cooling of the tube base shall be increased with higher values of bias or frequency. The approximate pressure drop values shown include the EIMAC SK-300A air-system socket or equivalent, with an EIMAC SK-1306 chimney, or equivalent, with air passing in a base-to-anode direction. Airflow shall be applied before or simultaneously with the application of electrode voltages (including the filament), and should normally be maintained for a brief period after all voltages are removed to allow for tube cool-down. In cases where long life and consistent performance are factors, cooling in excess of minimum requirements is normally beneficial.

TABLE I. Minimum airflow requirements.

Anode dissipation (watts)	Airflow (cfm)	Approximate pressure drop (In. H_2O)
4,000	110	0.4
6,000	200	0.8
8,000	315	1.7
10,000	445	2.8
12,000	600	4.4

2. Any change in filament current resulting from the vibration or shock testing (considered individually) shall not exceed the specified limit for ΔIf.

3. In all electrical tests involving application of filament voltage, it is permissible to use an air-system socket and chimney, and forced-air cooling is permitted.

4. The tube manufacturer warrants the tube for one year from the date of shipment, or 1,000 hours of filament life, whichever first elapses. This warranty applies only when the tube is operated within the maximum ratings (see "Absolute ratings" of MIL-E-1). A defective tube shall either be replaced or at the option of the manufacturer a credit shall be made in the amount of the original purchase price prorated on the basis of 1,000 hours of "filament-on" time.

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Custodians:
Army - EL
Air Force - 80

Review activities:
Army - MU
Air Force - 11, 17
DSA - ES

User activities:
Army - WC
Air Force - 19

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2577)

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discarded during shall be mounted in a metal cell which shall not become

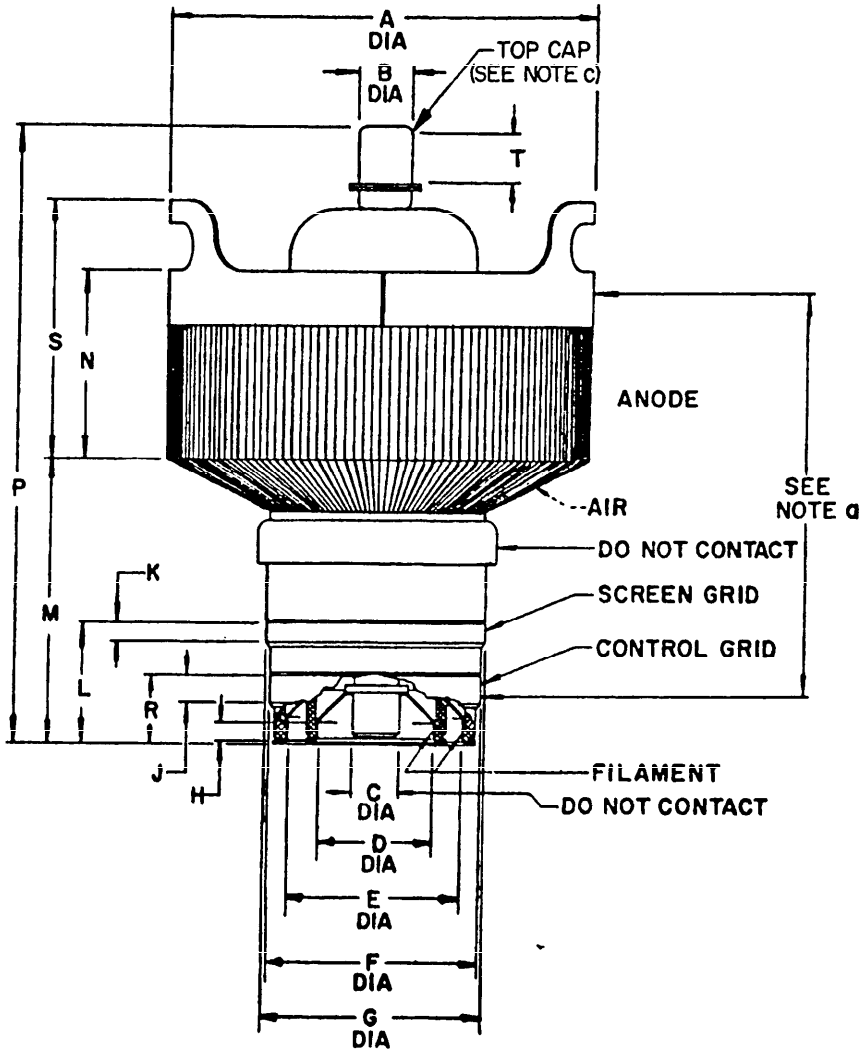


FIGURE 1. Outline drawing of electron tube type 8171.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
C	.720 (18.29)	.760 (19.30)
D	1.896 (48.16)	1.936 (49.17)
E	3.133 (79.58)	3.173 (80.59)
F	3.792 (96.32)	3.832 (97.33)
G	3.980 (101.09)	4.020 (102.11)
H	.188 (4.78)	
J	.188 (4.78)	
K	.188 (4.78)	
L	1.764 (44.81)	1.826 (46.38)
P	8.625 (219.08)	9.125 (231.78)
R	.986 (25.04)	1.050 (26.67)
T	.375 (9.53)	
Quality conformance inspection, part 3 (periodic check)		
A	6.928 (175.97)	7.050 (179.07)
B	.855 (21.72)	.895 (22.73)
M	4.186 (106.32)	4.568 (116.03)
N	2.412 (61.26)	2.788 (70.82)
S	3.412 (86.66)	3.788 (96.22)

NOTES:
a. The total indicator reading (the sum of the positive and negative deflection shown by the indicator when measuring the eccentricity of the surface with respect to another, with the reference axis established) of the screen grid and filament contact surfaces shall not exceed .040(1.02 mm) with respect to the control grid and anode contact surfaces when the latter surfaces are rotated on rollers at the points indicated by the arrows. (Quality conformance inspection, part 2 shall apply.)
b. Letters H, J, K, N and T also represents contact surfaces.
c. Top cap outline optional provided it meets requirements of dimensions B and T.

FIGURE 1. Outline drawing of electron tube type 8171 - Continued.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/1716 ELECTRON TUBE, POWER TYPE 8171

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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ESC-FM 1068-68

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Fort Monmouth, New Jersey 07703

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